

Module 3: Syntax Analysis (Parsing)**Question 5 (3 marks each):**

- **A-5:** What are viable prefixes?
- **B-5:** Define an operator grammar. Give an example.
- **D-5:** Explain synthesized attributes and inherited attributes.
- **E-5:** Define operator grammar. Construct an operator precedence table for the grammar:
 $E \rightarrow E + E \mid E * E \mid id.$
- **F-5:** Define an operator grammar. Give an example.
- **July 2023-5:** Differentiate CLR and LALR parsers.

Question 6 (3 marks each):

- **A-6:** What are the different parsing conflicts in the SLR parsing table?
- **B-6:** What are viable prefixes?
- **D-6:** Check whether the translation scheme $A \rightarrow B \{B.i = C.s\} C \{C.i = A.i\}$ is L-attributed or not. Justify.
- **E-6:** What are the different operations in a shift-reduce parser?
- **F-6:** What are the possible actions of a shift-reduce parser?
- **July 2023-6:** What are the possible actions of a shift-reduce parser?

Question 15 (14 marks each):

- **A-15:**
 - a) Construct a canonical LR(0) collection of items for the grammar:
 $S \rightarrow L = R \mid R, L \rightarrow *R \mid id, R \rightarrow L.$ Prove that this grammar is not SLR(1).
 - b) What is handle pruning? Indicate the handles in the reduction of the sentence `aaabbb` to the start symbol using the grammar:
 $S \rightarrow aABb, A \rightarrow aA \mid a, B \rightarrow bB \mid b.$
- **B-15:**
 - a) Construct a canonical LR(0) collection of items for the grammar:
 $S \rightarrow L = R \mid R, L \rightarrow *R \mid id, R \rightarrow L.$ Also, identify a shift-reduce conflict in the LR(0) collection.
 - b) Write an algorithm for computing the closure of LR(0) items.
- **D-15:**
 - a) Illustrate actions according to an operator precedence parser for the input `id1 id2 * id3` based on the grammar:
 $E \rightarrow E + E \mid E * E \mid (E) \mid id.$
 - b) Explain how to construct an operator grammar without ambiguity.
- **E-15:**
 - a) Construct an SLR parsing table for the grammar:
 $E \rightarrow T + E \mid T, F \rightarrow id.$ Check if the grammar is SLR or not. Justify.
 - b) What are handle and handle pruning? Indicate the handles in the reduction of the sentence `bbaaab` to the start symbol using the grammar:
 $S \rightarrow aB \mid bA, A \rightarrow a \mid aS \mid bAA, B \rightarrow b \mid bS \mid aBB.$
- **F-15:**
 - a) Construct a canonical LR(1) parsing table for the grammar:
 $S \rightarrow CC, C \rightarrow aC, C \rightarrow d.$
 - b) Illustrate the actions of the operator precedence parser for the input `id1 + id2 * id3` based on the grammar:
 $E \rightarrow E + E \mid E * E \mid (E) \mid id.$
- **July 2023-15:**
 - a) What is a shift-reduce parser? Explain in detail the conflicts that may occur during shift-reduce parsing.
 - b) Construct a canonical LR(1) collection of items for the grammar:
 $S \rightarrow CC, C \rightarrow aC, C \rightarrow d.$

Question 16 (14 marks each):

- **A-16:**
 - a) Derive an LR(1) parsing table for the grammar:
 $S \rightarrow Aa \mid bAc \mid Bc \mid bBa, A \rightarrow d, B \rightarrow d.$
 - b) Write all moves by the LR parser for parsing the input `bdc`.
- **B-16:**
 - a) Construct an LALR parse table for the grammar:
 $A \rightarrow BB, B \rightarrow bB \mid d.$
 - b) What are the different parsing conflicts in the SLR parsing table?
- **D-16:**
 - Construct a canonical LR(1) parsing table and perform parsing actions for a valid input according to the grammar:
 $S \rightarrow Aa \mid bAc \mid cAb, A \rightarrow a.$
- **E-16:**
 - a) Construct a CLR parsing table for the grammar:
 $S \rightarrow L = R \mid R, L \rightarrow *R \mid id, R \rightarrow L.$ Is the grammar CLR? Justify.
 - b) Write a note on different conflicts in a shift-reduce parser.
- **F-16:**
 - a) Explain the term handle and handle pruning with an example.
 - b) What is a shift-reduce parser? Explain the conflicts that may occur during shift-reduce parsing.
- **July 2023-16:**
 - Construct an LALR parsing table for the grammar:
 $S \rightarrow Aa \mid bAc \mid dc \mid bda, A \rightarrow d.$ Verify whether the input string `bdc` is accepted by the grammar.

Module 4: Semantic Analysis and Intermediate Code Generation

How can Grok help?

 DeepSearch  Think

Grok 3

